Unit-2
Lecture-1

de Broglie Hypothesis

- It states that –

There is a wave associated with every moving particle moving with velocity v, and the wavelength of this wave is given by –

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}}$$

$h = 6.626 \times 10^{-34}$ J-s is Plank's constant

$p = m\,v$ is momentum

de Broglie Hypothesis

Let us consider the case of the photon. Energy of the photon, according to Plank's theory of radiation is given by – $\boxed{E = h\nu}$ 1

where h is Plank's constant and ν is frequency of radiation

If we consider a photon as a particle of mass m, its energy is given by Einstein Mass Energy relation as – $\boxed{E = mc^2}$ 2

From equation (1) and (2), we get,

$$\boxed{h\nu = mc^2} \quad 3$$

As photon travels with velocity of light 'c' in free space, its momentum 'p' is given by –

$$\boxed{p = mc} \quad 4$$

de Broglie Hypothesis

$$E = h\nu \quad (1)$$

$$E = mc^2 \quad (2)$$

$$h\nu = mc^2 \quad (3)$$

$$p = mc \quad (4)$$

Dividing equation (3) by (4) we get –

$$\frac{h\nu}{p} = \frac{mc^2}{mc} = c$$

$$\therefore \frac{h}{p} = \frac{c}{\nu} = \lambda \qquad \left(\because \frac{c}{\nu} = \lambda\right)$$

$$\boxed{\therefore \lambda = \frac{h}{p}}$$

de Broglie wavelength in terms of Kinetic Energy

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}}$$ de Broglie Relation

The Kinetic energy of the particle is –

$$E = \frac{1}{2}mv^2 = \frac{1}{2m}m^2v^2 = \frac{1}{2m}p^2$$

$$\therefore p^2 = 2mE$$

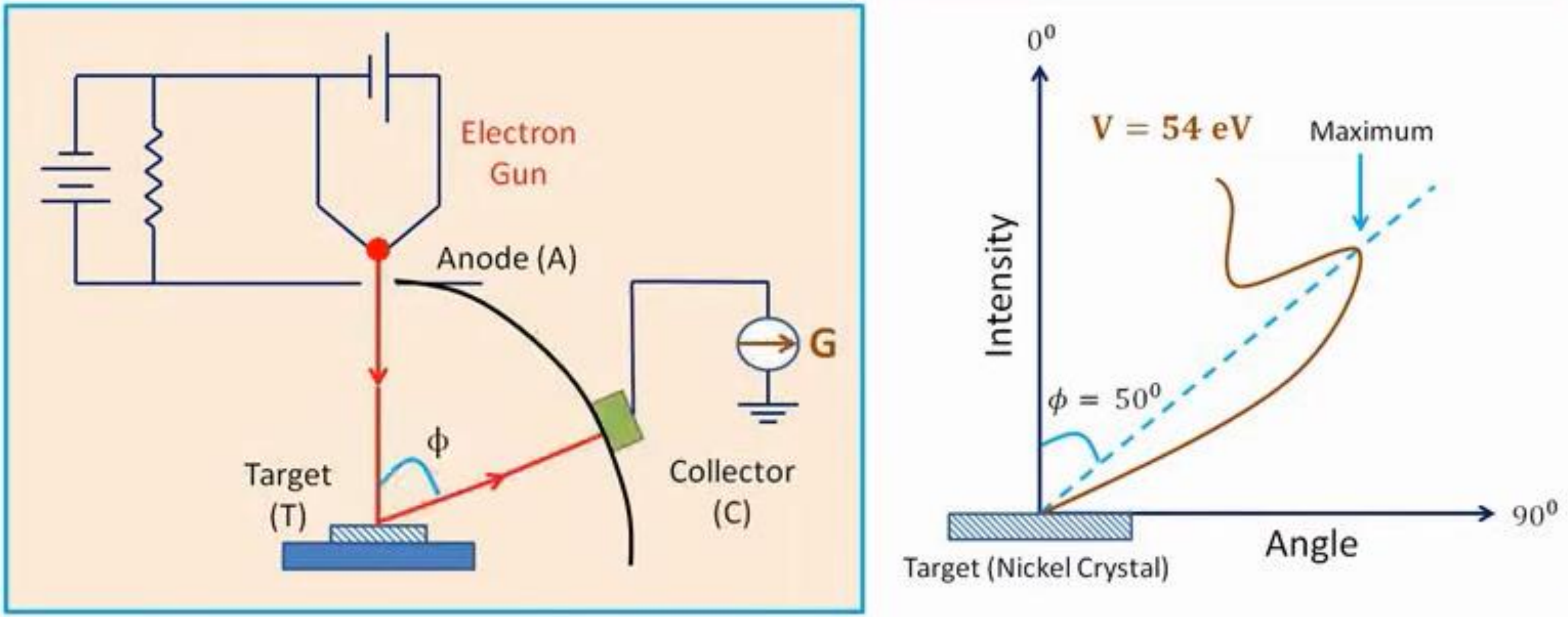
$$\therefore p = \sqrt{2mE}$$

$$\boxed{\therefore \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}}$$

$$\boxed{\therefore \lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}}$$

$$\boxed{\therefore \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mkT}}}$$

Experimental Verification of de Broglie Hypothesis (Davisson and Germer Experiment)

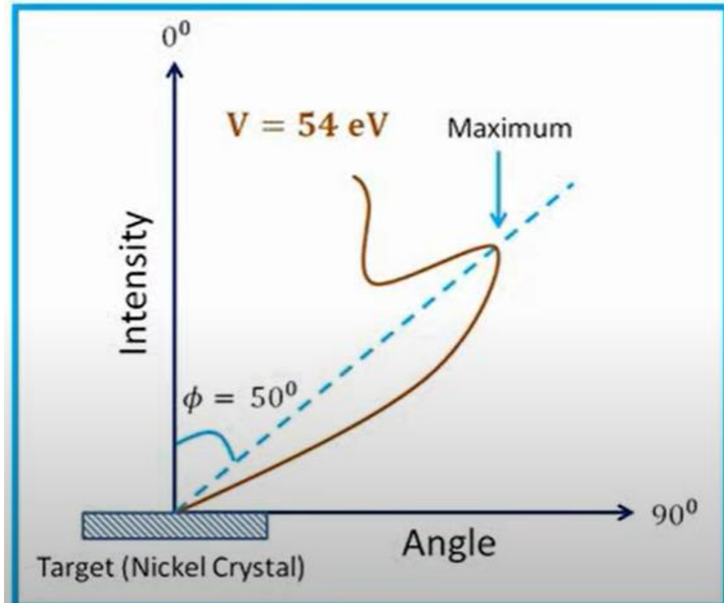


The first experimental evidence of de Broglie hypothesis came from the experiments performed by Davisson and Germer. Following figure shows the experimental arrangement used by Davisson and Germer.
The various constructional parts are shown in following fig. –

i) **Electron gun :** It consists of a tungsten filament F. Electrons are emitted by thermionic emission. Due to the action of electric field across the gun, a fine stream of electrons emerges out of this gun.
ii) **Anode (A) :** It accelerates the electrons towards the target.
iii) **Target (T) :** It is a single crystal of Nickel. It can be rotated about its axis, which is parallel to the axis of electron beam. The position of the crystal can be adjusted. A thin pencil beam of electrons is allowed to reflect from the crystal surface in different directions.
iv) **Collector (C) :** Collect the reflected electrons. It is a Faraday cylinder connected to a sensitive galvanometer G. It can be moved along the circular scale to locate the position of maxima and minima between the angle 20 and 90 degree. The inner and outer walls of the cylinder are insulated from each other and a retarding potential is applied between them so that only fastest moving electrons can enter the cylinder.
v) **Outer Chamber :** The whole arrangement is enclosed in an evacuated chamber.

In this experiment, the beam of electrons was made to fall normally on the surface of a crystal and then the collector (C) was moved to various positions on the scale. The galvanometer current for each position was noted. This current is a measure of the intensity of the diffracted beam of electrons. Graph of this current (intensity of diffracted beam) against the angle between the incident beam and the beam entering the collector was plotted. This procedure was repeated for different voltages and several curves were drawn as shown in following Fig.-

Experimental Verification of de Broglie Hypothesis (Davisson and Germer Experiment)



$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

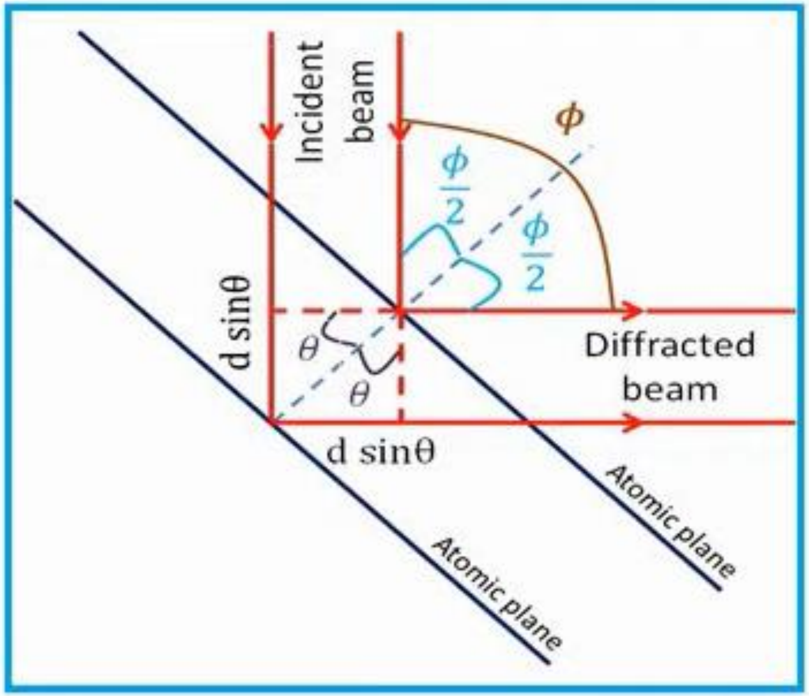
$$\therefore \lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 54}}$$

In one set of experiment, Davisson and Germer found that when the accelerating potential is raised to 54 volts, a maximum appears when the angle between incident and diffracted beam (= ϕ) was 50 degree for the first order spectrum. This maximum confirms the existence of diffraction phenomenon and hence the wavelike behavior of electrons. Hence, by using the de Broglie relation, the wavelength of the wave associated with the electron can be calculated and is given by $-\ h = 6.62\times10^{-34}$ Joules - sec, $m = 9.1\times10^{-31}$ Kg, $e = 1.6\times10^{-19}$ Coulomb, $V = 54$ volts

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2\,m\,e\,V}} = \frac{6.62\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1.6\times10^{-19}\times54}}$$

$$= 1.67\times10^{-10}\ \text{m} = 1.67\ \text{A}^0$$

Experimental Verification of de Broglie Hypothesis (Davisson and Germer Experiment)



From Bragg's diffraction law, we have

$$2\,d\sin\theta = n\lambda$$

$$\therefore \lambda = \frac{2\,d\sin\theta}{n}$$

$$\therefore \lambda = \frac{2 \times 9.086 \times 10^{-11} \times \sin 65}{1}$$

$$\boxed{\therefore \lambda = 1.65 \times 10^{-10}\,\mathrm{m} = 1.65\,\mathrm{A}^0}$$

This value matches with the wavelength calculated using de Broglie relation.

The wavelength of the wave can also be calculated by using Bragg"s relation which is established result in optics.

From Bragg's diffraction law, we have

$$2\,d\sin\theta = n\lambda$$

$$\therefore\ \lambda = \frac{2\,d\sin\theta}{n}$$

where $d = a\sin\phi = 2.15\times10^{-10}\times\sin 25^0 = 9.086\times10^{-11}$ is the inter - planar spacing

$a = 2.15\times10^{-10}$ m is the inter - atomic spacing

$\theta = 90^0 - \dfrac{\phi}{2} = 90^0 - 25^0 = 65^0$ is the angle between the diffracted beam and atomic planes.

$n = 1$ is the order of the spectrum

λ is the wavelength of the electron wave which is to be determined.

Substituting all these values in Bragg's law,

$$\lambda = \frac{2\,d\sin\theta}{n} = \frac{2\times9.086\,x\,10^{-11}\times\sin 65}{1} = 1.65\,A^0$$

This value matches with the wavelength calculated using de Broglie relation. This confirms the correctness of de Broglie relation.

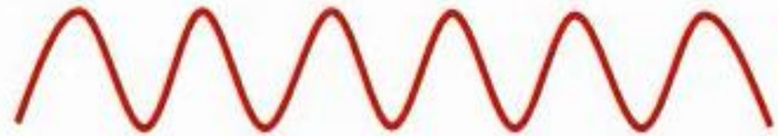
Heisenberg uncertainty principle

Properties of Matter Waves

- Waves associated with moving particles are called matter waves.
- Wavelength of matter wave is give by $\lambda = \frac{h}{p} = \frac{h}{mv}$
- Matter waves are not electromagnetic waves and can be associated with any particle whether charged or uncharged
- Matter waves can propagate in a vacuum, hence they are not mechanical wave.
- Phase velocity of matter wave $v_p = \frac{c^2}{v} > c$

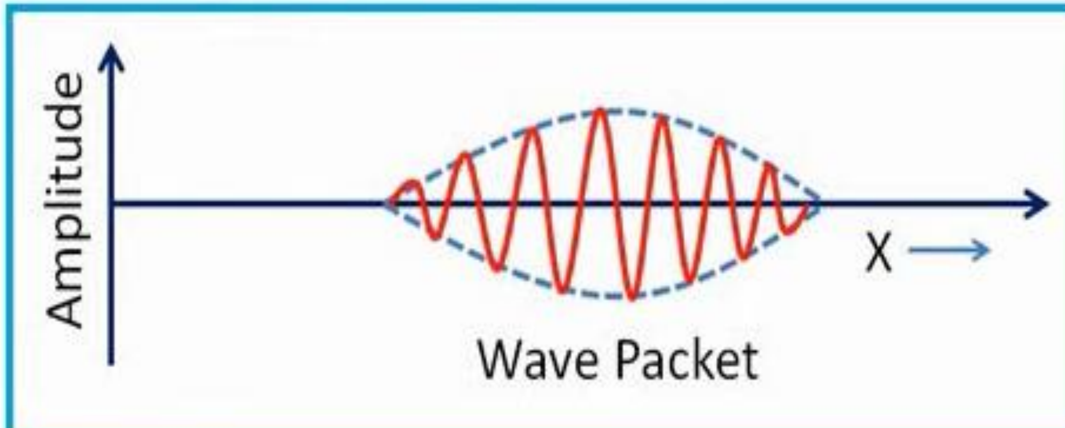
Which type of wave associated with matter?

- Single Monochromatic wave?



  - Location of the particle?
  - Phase velocity $v_p = \nu\lambda = \frac{h\nu\lambda}{h} = \frac{h\nu}{h/\lambda} = \frac{E}{p} = \frac{mc^2}{mv} = \frac{c^2}{v}$

- Schrodinger's solution
  - Group of waves
  - Wave packet



The **phase velocity** is the velocity with which a particular phase of the wave propagates in the medium.

Let the equation of the wave travelling in x-direction with vibrations in y-direction is –

$$y = A\sin(\omega t - kx) \quad (1)$$

Where A is amplitude of vibration,

$k = \frac{2\pi}{\lambda}$ is propagation constant,

$\omega = 2\pi\nu$ is the angular frequency

$$\therefore \lambda = \frac{2\pi}{k} \text{ and } \nu = \frac{\omega}{2\pi} \quad (2)$$

Phase velocity,

$$v_p = \nu\lambda = \frac{\omega}{2\pi} \times \frac{2\pi}{k} = \frac{\omega}{k} \quad (3)$$

From (2)

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/_{mv}} = \frac{2\pi mv}{h} \quad (4)$$

As from de Broglie relation says $\lambda = \frac{h}{mv}$

Phase Velocity

$$y = A\sin(\omega t - kx) \quad (1)$$

$$\lambda = \frac{2\pi}{k} \text{ and } \nu = \frac{\omega}{2\pi} \quad (2)$$

$$v_p = \nu\lambda = \frac{\omega}{2\pi} \times \frac{2\pi}{k} = \frac{\omega}{k} \quad (3)$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/_{mv}} = \frac{2\pi mv}{h} \quad (4)$$

Now, let us equate enegry $E = h\nu$ with energy $E = mc^2$

$$h\nu = mc^2 \Rightarrow \nu = \frac{mc^2}{h}$$

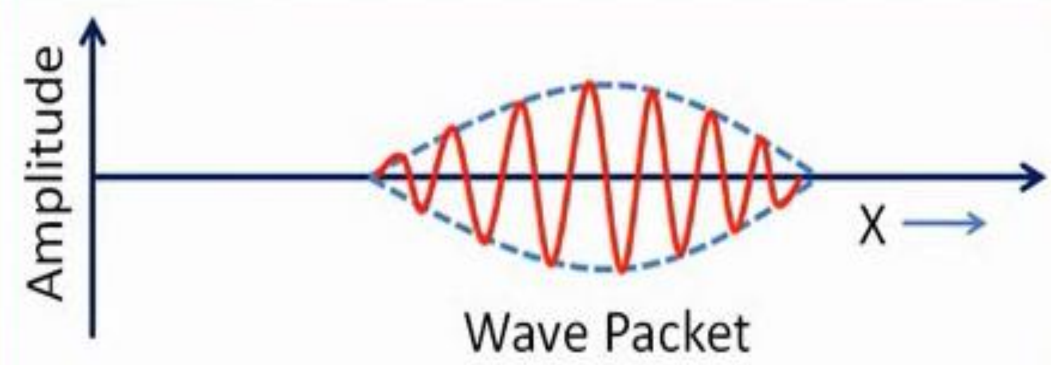
From (2) $\omega = 2\pi\nu = \frac{2\pi mc^2}{h}$ (5)

Phase velocity,

$$v_p = \frac{\omega}{k} = \frac{\frac{2\pi mc^2}{h}}{\frac{2\pi mv}{h}} = \frac{c^2}{v} \quad (6)$$

Schrodinger's solution

– Group of waves
– Wave packet



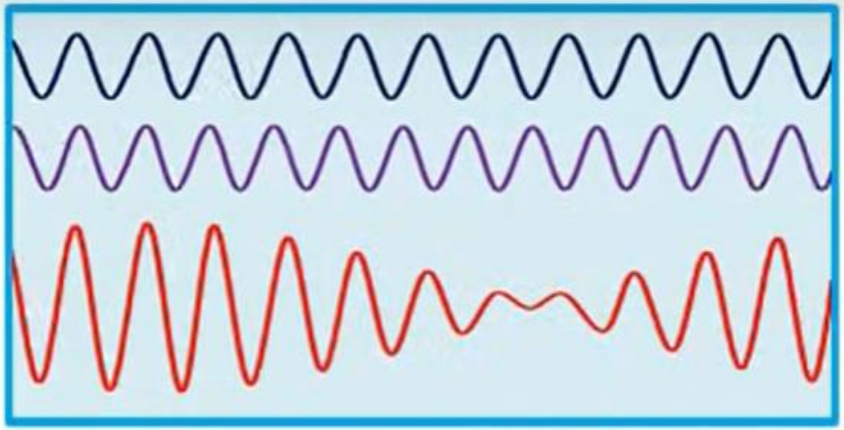
- ✓ The phases and amplitude of the waves in group are such that they undergo constructive interference only over a small region where the particle may be located.
- ✓ Outside this region, destructive interference occurs and amplitude is zero.
- ✓ Velocity with which this wave packet moves is called as group velocity. This group velocity is equal to the velocity of particle.

Group Velocity

Let us consider two waves represented by -

$y_1 = A\sin(\omega t - kx)$

$y_2 = A\sin[(\omega + d\omega)t - (k + dk)x]$



The resultant displacement y an any time t is $y = y_1 + y_2$

$$y = 2A\sin\left(\frac{2\omega + d\omega}{2}t - \frac{2k + dk}{2}x\right)\cos\left(\frac{d\omega}{2}t - \frac{dk}{2}x\right)$$

The sine term represents a wave of angular frequency ω and propagation constant k

The Cosine term modulates this wave with angular frequency $\frac{d\omega}{2}$ to produce wave-group travelling with velocity $v_g = \frac{d\omega}{dk}$

Group Velocity

We know - $\lambda = \frac{2\pi}{k}$ and $\nu = \frac{\omega}{2\pi}$ (1)

Group velocity $v_g = \frac{d\omega}{dk} = \frac{d\omega/_{dv}}{dk/_{dv}}$ (2)

$h\nu = mc^2 \Rightarrow \nu = \frac{mc^2}{h}$

$\omega = 2\pi\nu = \frac{2\pi mc^2}{h}$

Putting $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$ (Relativistic mass)

$$\omega = 2\pi\nu = \frac{2\pi c^2}{h}\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\therefore \frac{d\omega}{dv} = \frac{2\pi m_0}{h} v\left(1-\frac{v^2}{c^2}\right)^{-3/2} \quad (3)$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi mv}{h} = \frac{2\pi v}{h}\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\therefore \frac{dk}{dv} = \frac{2\pi m_0}{h}\left(1-\frac{v^2}{c^2}\right)^{-3/2} \quad (4)$$

Group Velocity

We know - $\lambda = \frac{2\pi}{k}$ and $\nu = \frac{\omega}{2\pi}$ (1)

Group velocity $v_g = \frac{d\omega}{dk} = \frac{d\omega/dv}{dk/dv}$ (2)

$$\therefore \frac{d\omega}{dv} = \frac{2\pi m_0}{h} v \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \quad (3)$$

$$\therefore \frac{dk}{dv} = \frac{2\pi m_0}{h} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \quad (4)$$

From (2) (3) and (4)

Group velocity

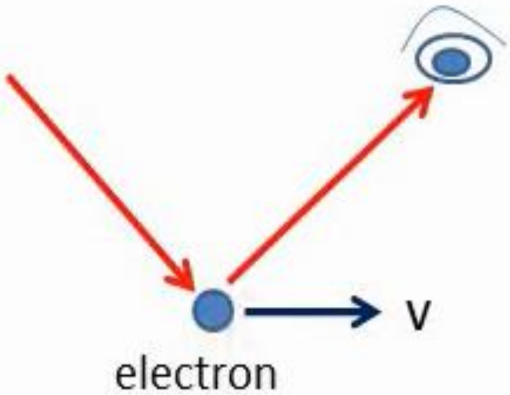
$$v_g = \frac{d\omega/dv}{dk/dv} = \frac{\frac{2\pi m_0}{h} v \left(1 - \frac{v^2}{c^2}\right)^{-3/2}}{\frac{2\pi m_0}{h}\left(1 - \frac{v^2}{c^2}\right)^{-3/2}}$$

$$\boxed{\therefore v_g = v}$$

Thus, group velocity associated with the wave packet is equal to the velocity of the particle.

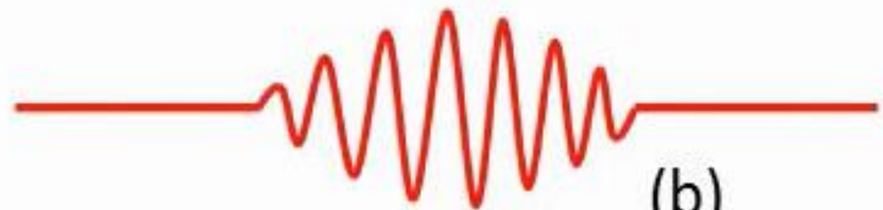
Heisenberg Uncertainty Principle

- If we are finding position of an moving electron, a photon must be seen by us after collision with the electron.
- But then the velocity and hence momentum of the electron will change due to collision.
- This will cause error or uncertainty in the measurement of momentum.
- If we try to determine momentum accurately, error or uncertainty in position is introduced.
- Thus, It is not possible to measure the position and momentum of an electron simultaneously and accurately.



Heisenberg Uncertainty Principle

(a)



(b)

- If the width of the wave packet is small as shown in fig. (a) then the particle can be located somewhat accurately, but the determination of wavelength (And hence the momentum) becomes a problem.
- If width of the wave packet is more (fig.(b)), then wavelength measurement (and hence determination of momentum) is accurate. However, position of the particle cannot be determined accurately.

Heisenberg Uncertainty Principle

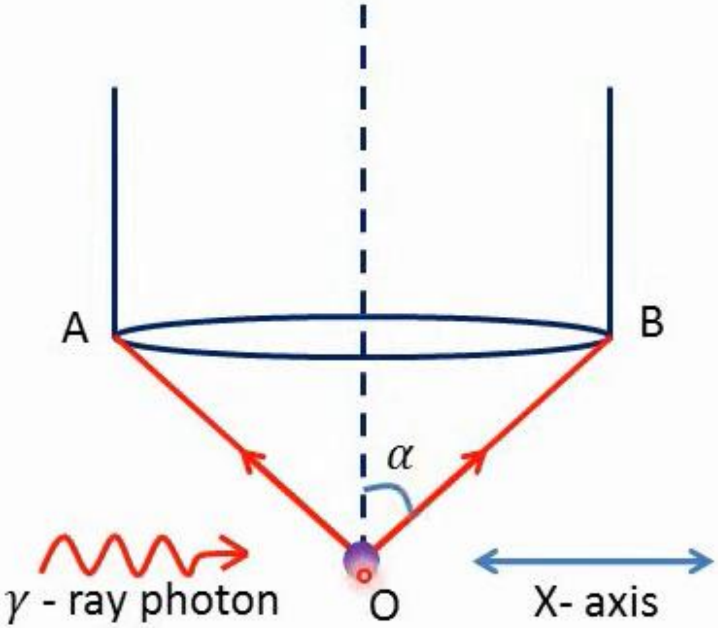
It is impossible to determine simultaneously, the position and momentum of the moving particle accurately. In any simultaneous determination of position and momentum of the particle, the product of uncertainties is equal to or greater than $\frac{\hbar}{2}$ i.e. $\frac{h}{4\pi}$

$$\boxed{\Delta x \ \Delta p_x \geq \frac{h}{4\pi}}$$

where Δx is uncertainty in measurement of position and

Δp_x is uncertainty in measurement of momentum along X-axis

(Experimental Verification of HUP)

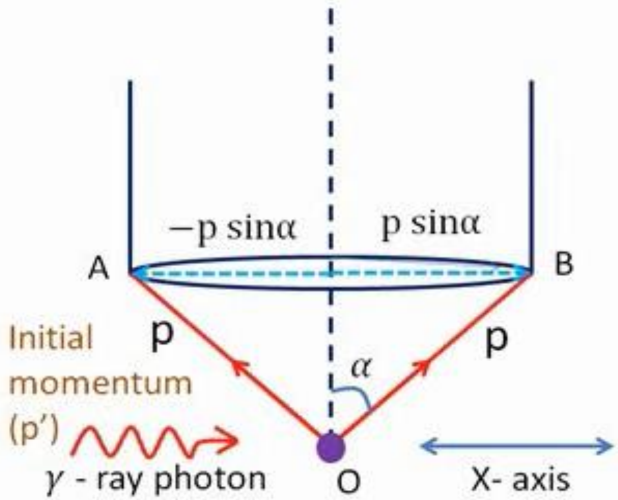


Heisenberg's Gamma Ray Microscope

Let the electron is at point 'O',

'λ' be the wavelength of γ-rays,

'α' be the semi-vertical angle of the cone of rays that enter the microscope objective.



Heisenberg's Gamma Ray Microscope

If scattered photon enters the objective along **OB**, its X-component of momentum would be $p \sin\alpha$

Hence, momentum imparted to the electron along X-axis would be $p' - p\sin\alpha$

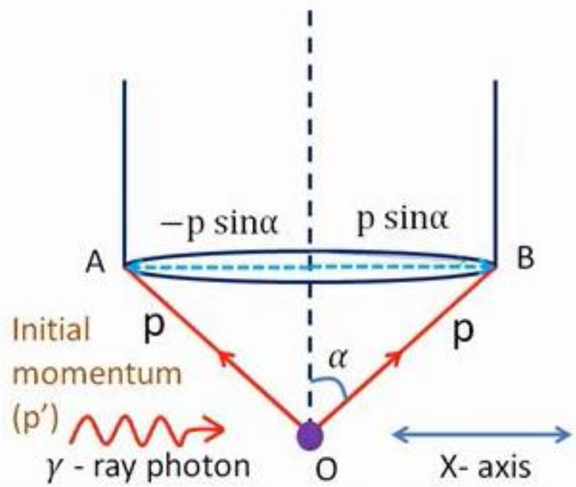
If scattered photon enters the objective along **OA**, its X-component of momentum would be $-p\sin\alpha$

Hence, momentum imparted to the electron along X-axis would be $p' - (-p\sin\alpha) = p' + p\sin\alpha$

Thus, uncertainty in momentum of an electron,

$$\Delta p_x = p' + p\sin\alpha - (p' - p\sin\alpha) = 2p\sin\alpha$$

(Experimental Verification of HUP)



Heisenberg's Gamma Ray Microscope

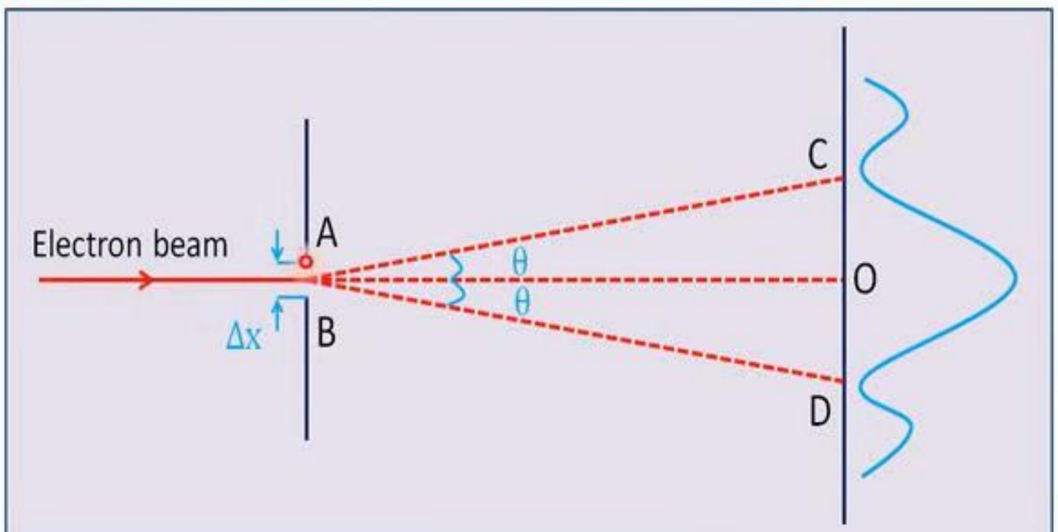
$$\Delta x = \frac{\lambda}{2\sin\alpha} \quad (1)$$

$$\Delta p_x = 2p\sin\alpha = 2\frac{h}{\lambda}\sin\alpha \quad (2)$$

$$\therefore \Delta x\,\Delta p_x = \frac{\lambda}{2\sin\alpha}2\frac{h}{\lambda}\sin\alpha$$

$$\boxed{\therefore \Delta x\,\Delta p_x = h > \frac{h}{4\pi}}$$

Single Slit Diffraction Experiment



Condition for minimum -

$d\sin\theta = n\lambda$

Here $d = \Delta x$

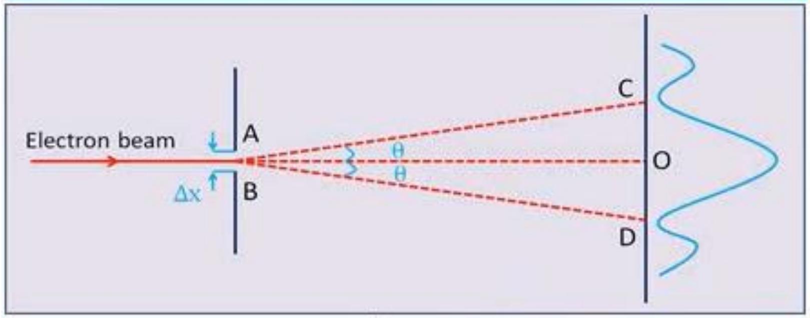
For 1st minimum, $n = 1$

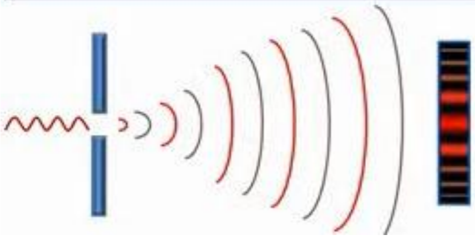
$\therefore \Delta x\sin\theta = 1\,\lambda$

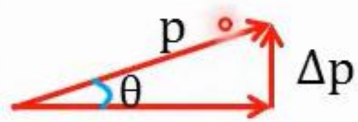
$$\boxed{\Delta x = \frac{\lambda}{\sin\theta}} \quad (1)$$

(Experimental Verification of HUP)

Single Slit Diffraction Experiment







$$\boxed{\Delta x = \frac{\lambda}{\sin\theta}} \quad (1)$$

$$\boxed{\Delta p = p\sin\theta} \quad (2)$$

$$\therefore \Delta x\,\Delta p = \frac{\lambda}{\sin\theta}\,p\sin\theta$$

$$\therefore \Delta x\,\Delta p = \frac{\lambda}{\sin\theta}\,\frac{h}{\lambda}\,\sin\theta$$

$$\boxed{\therefore \Delta x\,\Delta p = h > \frac{h}{4\pi}}$$

Heisenberg Uncertainty Principle

Heisenberg's Uncertainty Principle (HUP) is applicable to all conjugate or complimentary pairs of physical variables whose product has the dimension of Planck's constant 'h'. Some common such pairs are

- Energy-Time,
- Position-Linear momentum,
- Angular momentum-Angular displacement etc.

Time-Energy Uncertainty Principle

Let us consider a particle of mass 'm' moving with a velocity 'v' so that its K.E. is –

$$E = \frac{1}{2} m v^2$$

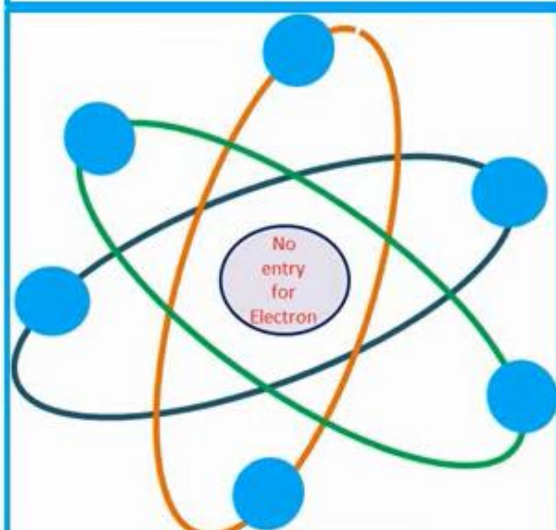
$$\therefore \Delta E = \frac{1}{2} m \, 2 \, v \, \Delta v \qquad \text{As } m\Delta v = \Delta p$$

$$= v \Delta p \qquad \text{As } \frac{\Delta x}{\Delta t} = v$$

$$= \frac{\Delta x}{\Delta t} \Delta p$$

$$\boxed{\therefore \Delta E \, \Delta t = \Delta x \, \Delta p \geq \frac{h}{4\pi}}$$

Why electron can not exist in nucleus?



Approximate radius of an nucleus $= r \approx 5 \times 10^{-15} \mathrm{m}$

$\therefore$Uncertainty in position if electron exist in nucleus $= \Delta x = 2r$

$$\therefore \Delta x = 2r \approx 10^{-14} \mathrm{m}$$

According to HUP $\boxed{\Delta x \, \Delta p \geq \frac{h}{4\pi}}$

$$\therefore \Delta x \, m\Delta v \geq \frac{h}{4\pi}$$

$$\therefore \Delta v \geq \frac{h}{4\pi \, m \, \Delta x} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 9.1 \times 10^{-31} \times 10^{-14}}$$

$$\therefore \Delta v \geq 5.797 \times 10^{9} \ \mathrm{m/s}$$

From this equation, the uncertainty of velocity is more than c ($3 \times 10^8 \mathrm{m/s}$). For this to happen, velocity of an electron must be greater that c. which is not possible. So the position of electron can't be in nucleus.